

#### 4. Penduga Bayes

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Klam → Rasio & Prior  
alternatif → Bayes

Bayes ← Prior  
Posterior (Prior + Evidence)

$$f(y|\theta) = \frac{f(y|\theta) \pi(\theta)}{\pi(\theta)}$$

Posterior ← Evidence

- 1) Likelihood →  $f_y(y|\theta)$
- 2) Joint:  $f(y, \theta) = f_y(y|\theta) \pi(\theta)$
- 3) Marginal Likelihood/Evidence  $f_y$
- 4) Posterior →  $\pi(\theta|y)$
- 5) Penduga Bayes

$X \sim \text{iid Bernoulli}(\theta) \rightarrow$  sukses (sukses)  $\rightarrow$  binomial  
 $Y = \sum_{i=1}^n x_i$  gunakan prior  $\theta \sim \text{Uniform}(0,1) \rightarrow \pi(\theta) = 1, 0 < \theta < 1$

$Y \sim \text{Binomial}(n, \theta) \rightarrow \text{prior } \pi(\theta) = 1$   
 $f_y(y|\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}, y = 0, 1, 2, \dots, n$

0) Likelihood  
 $f_y(y|\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$

1) Joint  
 $f(y, \theta) = f_y(y|\theta) \pi(\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y} \cdot 1$

2) Marginal  
 $f(y) = \int_0^1 f(y, \theta) d\theta = \int_0^1 \binom{n}{y} \theta^y (1-\theta)^{n-y} d\theta$

3) Prior  
 $\pi(\theta) = 1$

4) Posterior  
 $\pi(\theta|y) = \frac{f(y, \theta)}{f(y)} = \frac{\binom{n}{y} \theta^y (1-\theta)^{n-y}}{\int_0^1 \binom{n}{y} \theta^y (1-\theta)^{n-y} d\theta}$

5) Penduga Bayes  
 $\hat{\theta}_B = E(\theta|y) = \int_0^1 \theta \pi(\theta|y) d\theta$

Posterior  
 $\pi(\theta|y) = \frac{f(y, \theta)}{f(y)}$

Penduga Bayes  
 $\hat{\theta}_B = E(\theta|y) = \int_0^1 \theta \pi(\theta|y) d\theta$

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 $\hat{\theta}_B = E(\theta|y) = \int_0^1 \theta \pi(\theta|y) d\theta$

Sehingga  $\hat{\theta}_B = \frac{y+1}{n+2}$

$Y \sim \text{Binomial}(n, \theta) \rightarrow Y=y$   
prior  $\theta \sim \text{Beta}(\alpha, \beta)$

1) Likelihood  
 $f_y(y|\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$

2) Prior  
 $\pi(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$

3) Joint  
 $f(y, \theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y} \cdot \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}$

4) Posterior  
 $\pi(\theta|y) = \frac{f(y, \theta)}{f(y)} = \frac{\binom{n}{y} \theta^y (1-\theta)^{n-y} \cdot \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}}{\int_0^1 \binom{n}{y} \theta^y (1-\theta)^{n-y} \cdot \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1} d\theta}$

5) Penduga Bayes  
 $\hat{\theta}_B = E(\theta|y) = \int_0^1 \theta \pi(\theta|y) d\theta$

Posterior  
 $\pi(\theta|y) = \frac{f(y, \theta)}{f(y)}$

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 $\hat{\theta}_B = E(\theta|y) = \int_0^1 \theta \pi(\theta|y) d\theta$

Posterior  
 $\pi(\theta|y) = \frac{f(y, \theta)}{f(y)}$

Penduga Bayes  
 $\hat{\theta}_B = E(\theta|y) = \int_0^1 \theta \pi(\theta|y) d\theta$

$Y \sim \text{Poisson}(\lambda)$  dan  $I = \sum_{i=1}^n Y_i$

$\lambda \sim \text{Gamma}(\alpha, \beta)$   
Temukan bentuk dis. posterior  $\pi(\lambda|I)$   
Temukan mean posterior  $E(\lambda|I)$

1) Likelihood  
 $f_y(y|\lambda) = \frac{e^{-\lambda} \lambda^y}{y!}$

2) Prior  
 $\pi(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta \lambda}$

3) Joint  
 $f(y, \lambda) = f_y(y|\lambda) \pi(\lambda) = \frac{e^{-\lambda} \lambda^y}{y!} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta \lambda}$

4) Posterior  
 $\pi(\lambda|y) = \frac{f(y, \lambda)}{f(y)} = \frac{\frac{e^{-\lambda} \lambda^y}{y!} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta \lambda}}{\int_0^\infty \frac{e^{-\lambda} \lambda^y}{y!} \cdot \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta \lambda} d\lambda}$

5) Penduga Bayes  
 $\hat{\lambda}_B = E(\lambda|y) = \int_0^\infty \lambda \pi(\lambda|y) d\lambda$

Posterior  
 $\pi(\lambda|y) = \frac{f(y, \lambda)}{f(y)}$

Penduga Bayes  
 $\hat{\lambda}_B = E(\lambda|y) = \int_0^\infty \lambda \pi(\lambda|y) d\lambda$

Posterior  
 $\pi(\lambda|y) = \frac{f(y, \lambda)}{f(y)}$

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